Neighborhoods, Family and Intergenerational Mobility

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DISCUSSION PAPER

Institutt for samfunnsøkonomi
Department of Economics

SAM 06/26

ISSN: 0804-6824

May 2026

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To what extent do childhood neighborhoods shape long-run socio-economic outcomes, and through which mechanisms? Using the quasi-random assignment of refugee children across neighborhoods in Denmark, we show that exposure to higher-quality neighborhoods—as measured by average neighborhood income and the wage outcomes of permanent resident children—raises labor force participation and market income in adulthood. Beyond economic integration, better neighborhoods further promote social integration by increasing educational attainment and naturalization. Applying a causal mediation analysis, we reject full mediation via neighborhood and school characteristics but not via parental income, pointing to the family as a fundamental mediator of neighborhood effects.

JEL Codes: *R23, I24, J15, I38, J61*
Keywords: *neighborhood effects, intergenerational mobility, migrant integration, causal mediation, parental investments*

Understanding how childhood environments shape adult economic outcomes is a long-standing question in the social sciences. Identification is challenging due to non-random sorting of households across neighborhoods. Moreover, the channels through which neighborhoods matter are not well understood (Chyn and Katz, 2021). To overcome these concerns, we exploit random variation in the neighborhood assignment of refugees generated by the Danish Dispersal Policy (DDP), carried out from 1986 to 1998. Leveraging this variation, we estimate the causal effect of childhood exposure to better neighborhood quality on long-term labor market outcomes, educational attainment and citizenship acquisition. We investigate mechanisms leveraging recent advances in causal mediation analysis.

This paper builds on rich administrative data from Denmark, allowing us to track refugee children from birth through to age 30 and to link them to their parental characteristics. We observe both the quasi-randomly assigned neighborhood and the long-run socio-economic outcomes of these children. We measure neighborhood quality in two ways. First, we average neighborhood income across residents at assignment. Second, following Chetty and Hendren (2018*b*), we esti-

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mate the income rank of permanent residents conditional on their parental income rank, for each neighborhood and cohort.[1]

Two main results emerge. First, childhood assignment to higher-quality neighborhoods raises long-run socio-economic outcomes. Exploiting the quasi-random assignment of refugees to Danish neighborhoods, we show that the quality of the assigned neighborhood positively and significantly impacts children's income, employment and reduces government transfers.[2] Assignment to a neighborhood where the mean income rank of children of permanent residents is 1 percentile higher (at a given level of parental income) increases children's income rank in adulthood by .5 percentiles. In other words, children of refugees placed in better quality neighborhoods pick up 50% of the difference in permanent residents' outcomes associated with the assigned neighborhood. Beyond economic integration, we extend the literature to provide novel evidence that better neighborhoods deepen social integration: a ten percentile increase in neighborhood quality raises education by 3 months and Danish citizenship acquisition by 2.5 percentage points.

Second, we introduce a causal mediation framework and find evidence consistent with intergenerational neighborhood effects operating through family inputs. This analysis is motivated by the statistically significant effects of better neighborhoods on parental income.[3] Our approach to mediation analysis imposes weak assumptions to identify the effect of a mediator on outcomes, based on recent work by Kwon and Roth (2024). Through our mediation analysis, we find that we cannot reject the sharp null of full mediation of neighborhood effects through parental income. We test and reject the sharp null of full mediation for several neighborhood characteristics, including school peer quality. These findings shed light on the nature of long-run neighborhood effects and the production of human capital. Exposure to better neighborhood quality directly fosters considerable economic integration among first-generation migrants. Thus, neighborhood quality contributes to improved long-term socio-economic outcomes of refugee children by raising parental input in the human capital production function. Taken together, the mechanisms through which neighborhood effects shape long-term outcomes differ between parents and children.

We contribute to the social science literature on whether—and how—childhood neighborhoods shape adult outcomes. Descriptive analyses (e.g., Wilson, 1987; Jencks and Mayer, 1990; Sampson, Morenoff and Gannon-Rowley, 2002), exper-

[1]This is akin to an observational value-added estimate of neighborhood quality.

[2]This latter finding adds to a mixed evidence base on neighborhood quality and welfare receipt (Bertrand, Luttmer and Mullainathan, 2000; Chyn, 2018; Kawano et al., 2024).

[3]Although prior evidence on the effects of neighborhood quality on adult outcomes is mixed, our results are consistent with a number of papers, (e.g., Edin, Fredriksson and Åslund, 2003; Pinto, 2021; Aliprantis and Richter, 2020) that find positive effects. Leveraging the DDP, Damm (2014) shows that higher co-ethnic employment in the assigned neighborhood improves labor-market outcomes, but does not find positive effects of neighborhood quality on parental outcomes. Using a distinct neighborhood quality measure and longer term panels, we find stronger and statistically significant neighborhood effects on adult outcomes.

imental evidence from voucher programs (e.g., Kling, Liebman and Katz, 2007; Ludwig et al., 2013; Chetty, Hendren and Katz, 2016; Aliprantis and Richter, 2020), quasi-experimental evidence (e.g., Chyn, 2018; Haltiwanger et al., 2024; Nakamura, Sigurdsson and Steinsson, 2022; Kawano et al., 2024) and mover designs with administrative data document the importance of exposure to better neighborhoods on the long-run adult outcomes (e.g., Chetty, Friedman and Rockoff, 2014; Chetty and Hendren, 2018*a*,*b*; Corak, 2020; Alesina et al., 2021; Deutscher, 2020; Card, Rothstein and Yi, 2025). Building on this literature, recent work provides evidence that non-compliance in voucher programs, sorting into neighborhoods and sampling error can bias estimates of neighborhood effects (Pinto, 2021; Eshaghnia, 2021; Cholli et al., 2024; Mogstad et al., 2024; Andrews, Kitagawa and McCloskey, 2024). We extend the literature in three main ways.

First, we identify neighborhood effects from random assignment to destinations—rather than voucher lotteries or displacement shocks—a design used in few papers to date.[4] This approach circumvents concerns in the identification of neighborhood effects arising from non-compliance or neighborhood sorting, to deliver estimates of the effect of childhood neighborhood quality on a range of key socioeconomic outcomes. Moreover, because all families are moving, we avoid confounding relocation effects with changes in neighborhood environment.

Second, we embed this design in a causal mediation framework to study mechanisms. This approach provides the first causal evidence that long-term neighborhood effects can operate largely through family channels versus place-based amenities.[5] This evidence is based on a unique population—migrants. While external validity warrants caution, findings from Boustan et al. (2025) investigating mobility in 15 destination countries, show that migrant and native intergenerational income transmission follows similar patterns, particularly in Denmark.[6] Second-generation gaps between natives and migrants are driven largely by parental income gaps—about one-half on average and almost 80% in Denmark—consistent with evidence from our causal mediation analysis.

Third, we investigate the role of neighborhoods in fostering the integration of second-generation migrants. While a number of studies analyze neighborhood effects using the DDP, this paper is the first, to the best of our knowledge, to examine intergenerational neighborhood effects on refugees' children measured

[4] Kawano et al. (2024) uses variation in random placement of military families and Damm and Dustmann (2014) uses variation from the DDP to estimate the effects of neighborhood crime on young-adult outcomes.

[5] Using a variance decomposition framework, prior work has emphasized that family background may play a larger role than neighborhoods in shaping long-run outcomes (see, for instance, (Solon, Page and Duncan, 2000; Oreopoulos, 2003; Raaum, Salvanes and Sørensen, 2006)). Agostinelli et al. (2019) also identify a family channel in a structural model with outcomes measured in adolescence. Moreover, in line with our results, Wodtke et al. (2023) find that schools do not mediate neighborhood effects. Statistical mediation analyses in the context of the DDP provide further evidence consistent with the family channel mediating neighborhood effects (Eshaghnia and Razavi, 2025).

[6] See also Jensen and Manning (2025) for a similar finding on the role of parental income for second-generation migrant outcomes in Denmark.

around age 30.[7] Our work goes beyond economic outcomes—income, employment and government transfers—to examine social integration—by investigating effects on educational attainment and naturalization. It informs the design of refugee assignment policies by highlighting their intergenerational consequences (Delacrétaz, Kominers and Teytelboym, 2023).

This paper unfolds as follows. Section I discusses the DDP. Section II describes our data. Section III presents our empirical strategy and neighborhood sorting patterns. Section IV presents our main estimates of long-term neighborhood effects. Section V describes our mediation analysis framework and provides results. Section VI evaluates the sensitivity of our results to different measures of neighborhood quality. Section VII concludes.

I. Dispersal Policy In Denmark

Denmark is one of several European countries that has adopted a refugee dispersal policy. As in many other European countries, the number of refugees increased sharply in Denmark during the 1980s. In this paper, we focus on the DDP, which was in place between 1986 and 1998.

Prior to the reform in 1986, refugees were allowed to choose where to settle, which led to a high concentration of refugees in larger cities.[8] A national dispersal policy implemented in 1986 aimed to distribute refugees more evenly across spatial units so that the costs of integration of migrants could be more evenly allocated across municipalities. The dispersal policy succeeded in achieving a more dispersed geographical distribution of refugees.[9]

[7]Recent studies have exploited the DDP to analyze the role of neighborhood characteristics on refugee outcomes such as labor market outcomes (Damm (2014)); Damm (2009); Damm and Rosholm (2010); Azlor, Damm and Schultz-Nielsen (2020); Eckert, Hejlesen and Walsh (2022)), labor market integration (Arendt (2022)), health (Hasager and Jørgensen (2021); Foverskov et al. (2022); Kim et al. (2023)), participation in crime (Damm and Dustmann (2014)); academic outcomes (Damm et al., 2024; Jensen, 2024), and preference formation (Abeler, Fosgaard and Gårn Hansen, 2023). Another strand of literature has used the DDP to study the impact of refugees on natives' behavior and natives' outcomes, including voting behavior (Dustmann, Vasiljeva and Piil Damm (2019)) and skill acquisition (Foged and Peri (2016)). Previous papers have studied refugee dispersal policies in other countries. See Edin, Fredriksson and Åslund (2003) for the causal effect on labor market outcomes of living in enclaves in Sweden; Chakraborty and Schüller (2022) for a similar analysis across numerous countries; Andersen, Osland and Zhang (2023) for integration of refugees in Norway; Kristiansen et al. (2022) for the impact of neighborhood labor market characteristics on refugee employment outcomes in the Netherlands; Vogiazides and Mondani (2021) for determinants of inter-regional immobility of refugees in Sweden; Robinson, Andersson and Musterd (2003) for an analysis of the dispersal policies in the Netherlands, Sweden, and the UK; Ravn, Österberg and Thomsen (2022) for the labor market policies for refugees in Denmark, Sweden, the Netherlands, and Germany. Brell, Dustmann and Preston (2020) provide an overview of refugee labor market integration outcomes across nine different high-income countries. See Nielsen Arendt, Dustmann and Ku (2022) for a review of the literature.

[8]There have been heated debates about immigration policies among policy makers in many countries. In some Western European countries, policies around asylum seekers and refugees were politicized, where asylum seekers and refugees were perceived by some political parties as a burden. Some Western European governments used various policies to spatially "spread the burden" by dispersing ethnic minorities, especially asylum seekers and refugees (see Robinson, Andersson and Musterd (2003) and Wren (2003)).

[9]Damm and Dustmann (2014) report that after 8 years, one in two households still lived in the area of initial assignment. Wren (2003) studies if the dispersal policy in Denmark has achieved its goals at macro vs micro levels. She argues that while, at a national level, the policy has achieved its objectives, at a

Damm (2005) investigates whether the DDP carried out from 1986 to 1998 on new refugees can be regarded as a natural experiment and whether refugees were randomly assigned to a location. She documents that the actual settlement has been influenced by five refugee characteristics and concludes that the initial location of new refugees 1986-1998 may be regarded as random when controlling for age, origin, year of assignment, household size, and marital status—all of which are observable.

II. Data and Measures of Neighborhood Quality, Outcomes and Mediators

A. Main Samples

Our analysis sample comprises all individuals born between 1986 and 1991 whose parents were assigned quasi-randomly to neighborhoods by the DDP. Narrowing down to these birth cohorts allows us to observe their later-life outcomes at around age 30.

B. Neighborhood Quality

This paper is based on two distinct measures of neighborhood quality. First, neighborhood quality is computed as the average household wage income, pre-tax and pre-transfer, at the municipality level. We then rank municipalities by this measure into percentiles, computed at the time of refugee assignment. We use the municipality as the neighborhood unit.[10] This is the level at which randomization takes place.

Second, following Chetty and Hendren (2018*b*), we construct a complementary measure of neighborhood quality, which we refer to throughout as an (observational) value-added measure. Using the sample of native Danes, we compute the expected income rank of children of permanent residents (at age 30) in each municipality, conditional on their parental income rank and birth cohort.

Formally, let y_i denote the child's adult income percentile rank based on their position in the national income distribution relative to all others in her birth cohort. Moreover, for child i, let $p(i)$ be the parent percentile rank in the national distribution of parental income for child i's birth cohort. Now, let $\bar{y}_{pks}$ denote the mean rank of children with parents at percentile p of the income distribution, residing in the neighborhood k for the entire childhood stage (ages 0 to 18), and birth cohort s. The mean child rank, given the rank of the parents in each neighborhood k and the birth cohort s is approximated by a linear form:

$$y_i = \alpha_{ks} + \psi_{ks} p_i + \epsilon_i \,. \tag{1}$$

regional scale the policy has effectively resulted in socio-ethnic spatial segregation in areas experiencing pre-existing deprivation and social exclusion.

[10] There were 271 municipalities in Denmark during the sample years (i.e., 1982-2000). These 271 municipalities were merged into 98 large new municipalities in 2007.

We obtain estimates of $\bar{y}_{pks}$ from the fitted values of the linear regression:

$$w_{kp} \equiv \bar{y}_{pks} = \hat{\alpha}_{ks} + \hat{\psi}_{ks} p \,. \tag{2}$$

We obtain estimates of the intercept α and slope ψ for each neighborhood using the sample of permanent resident Danes. Then, for a given refugee family with parental income rank p, we compute the quality of the assigned neighborhood (w_{kp}) from Equation 2. We measure parental income rank p between the ages 0–5 of the child, within the birth cohort of children.

C. Outcome Variables and Mediators

We describe the different outcomes and mediators we analyze. See Appendix S2 for more details on the construction of variables.

Economic Outcomes We focus on two definitions of children's adult economic outcomes. *Pre-tax* income captures wage income excluding transfers and taxes to focus on earnings potential. This income measure is contrasted with a *post-tax* income measure, capturing annual net income, to account for the significant tax and transfer system in Denmark (Landersø and Heckman, 2017; Eshaghnia et al., 2022). Finally, we also study the extensive margin by analyzing employment status as recorded in the administrative registers. We average these child income measures across ages 29 and 30.

Years of Education To capture human capital accumulation, we further compute a measure of individuals' years of education, by converting the degree completed to the corresponding years of completion. This measure is captured at age 29 of the child.

Danish Citizenship We define an indicator equal to one if the individual has Danish citizenship at age 30, and zero otherwise.

Our analysis further investigates the role of mediators. We measure the parental mediator—parental household income—as cumulative parents' pre-tax income during the child's ages 0–18 to approximate permanent income.[11] We measure neighborhood characteristics—including local crime rates, average years of education and share of migrants—between ages 0 and 18. The exception is peer middle school grades (9th grade) at neighborhood-level, which are only available from ages 15–18.[12]

Tables S1 and S2 in the Appendix report summary statistics for our sample, respectively in levels and ranks.

[11] We run our analyses measuring parental income between ages 0–30 and find similar results.
[12] School grades are only available from 2001 onward.

III. Childhood Exposure to Neighborhood Quality and Long-Term Outcomes

A. *Empirical Strategy*

In this paper, we relate later-life outcomes of children to the assigned neighborhood quality, through the following empirical specification:

$$y_i = \alpha + \beta w_i + \boldsymbol{\gamma'} \boldsymbol{x}_i + \varepsilon_i \tag{3}$$

where the variable y_i denotes the later-life outcomes we presented in Section II for individual i measured at around age 30. Neighborhood quality is denoted as w_i, measured at the time of assignment. $\boldsymbol{x_i}$ denotes a vector of variables, used by the authority to place refugees, including the age of the parents, year of assignment, country of origin, household size, and marital status, as motivated by the work of Damm (2005).

Given the quasi-random allocation of refugees to neighborhoods, neighborhood quality at assignment is uncorrelated with ε_i in Equation (3), conditional on $\boldsymbol{x_i}$. β can therefore be interpreted as an intent-to-treat (ITT) estimate of the effect of assignment to a better quality neighborhood on later-life outcomes.

B. *Sorting Patterns*

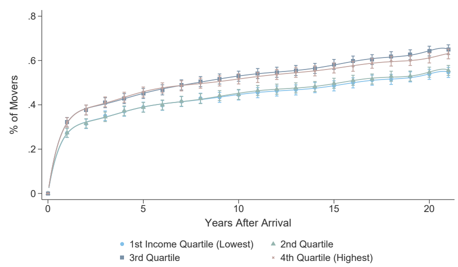
To interpret the ITT captured by β in Equation 3, we examine the mobility patterns of refugees from their initially assigned neighborhoods over time. Panel (a) of Figure 1 shows that even after ten years, more than 50% remain in their original location, with slightly lower persistence if assigned to a higher quality neighborhood.

Despite high mobility rates, panels (b) and (c) of Figure 1 demonstrate that movers typically select neighborhoods of similar quality over time, though regression to the mean becomes apparent after 10–15 years using the average neighborhood income measure.[13]

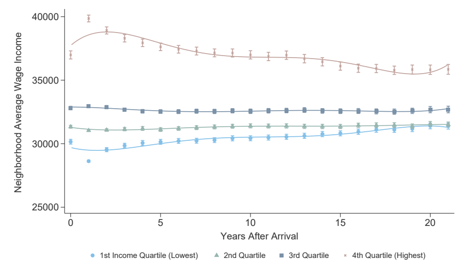
The upshot from analyzing neighborhood sorting patterns is that the ITT estimates recovered by Equation 3 in fact reflect the effects to the neighborhood exposed during childhood on long-term outcomes. It is worth noting that this estimand differs from that of a number of previous studies on neighborhood effects relying on the Moving to Opportunity (MTO) experiment. In the MTO setting, where individuals are randomly provided voucher subsidies to help disadvantaged families move to better neighborhoods (Katz, Kling and Liebman (2001); Goering, Feins et al. (2003); Kling, Liebman and Katz (2007); Gennetian et al. (2012)),

[13] As documented in Online Appendix Section S3, the result holds across multiple definitions of neighborhood quality. Figure S1 reports time series based on the Equation 2 definition (with gross and net income) and on time-varying neighborhood characteristics, including average neighborhood income and average education.

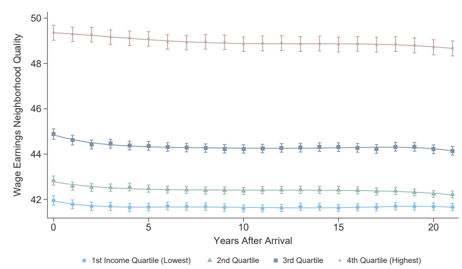
Figure 1. Sorting of Individuals by Initial Neighborhood Quality of Assignment



Panel A. Percentage of Movers



Panel B. Neighborhood Quality based on Wage Income Rank



Panel C. Neighborhood Quality based on Neighborhood Value-Added

Note: Panel (a) reports percentage of movers to a different municipality over time. Panel (b) reports neighborhood quality over time, as defined by our average wage income measure. Panel (c) replicates the analysis using our neighborhood value-added measure, as defined in Equation 2. We split households into quartiles based on the assigned neighborhood quality. Using the `binsreg` command in `Stata`, a global polynomial regression model was fitted for plotting purposes. The confidence intervals were constructed using a piecewise polynomial of degree 1 with one smoothness constraint.

the ITT refers to the effect of the voucher. Nearly half of the families did not redeem the vouchers. This additionally leads to challenges in the identification of neighborhood effects when scaling ITT estimated by the first stage compliance rate to obtain the LATE.[14] In contrast, DDP addresses self-selection from two margins: neighborhood placement is binding, and subsequent moves did not materially alter refugee children's neighborhood quality exposure relative to their initial assignments.

IV. Results

A. *Children Long-term Outcomes*

Table 1 presents estimates of neighborhood effects on economic and social integration outcomes, using our two distinct measures of neighborhood quality:

[14]See Pinto (2018) for a more detailed discussion of this issue.

average neighborhood income, in rank, (columns 1–2) and neighborhood (observational) value-added (columns 3–4). Across both panels, we find consistent evidence that exposure to higher-quality neighborhoods improves long-run outcomes.

In Panel A, a ten percentile increase in neighborhood quality leads to a .45 and .35 percentile increase in pre-tax and post-tax income, a 1 percentage point increase in employment probability, and a $140 reductions in yearly government transfers. This evidence suggests that neighborhood environments shape not only earnings capacity but also reliance on the social safety net.

Focusing on our value-added measure in Column (4), we find that assignment to a neighborhood where the mean income rank of children of permanent residents is 1 percentile higher (at a given level of parental income) increases children's income rank in adulthood by approximately .50 percentiles. In other words, children of refugees placed in better quality neighborhoods pick up 50% of the difference in permanent residents' outcomes associated with the assigned neighborhood. β can also be thought of as a forecasting coefficient (Abaluck et al., 2021).[15] We find similar effects on post-tax income of children, with a coefficient of .45, as shown in column (4) of Table 1.

Panel B extends these findings to social integration, showing that neighborhood quality positively impacts both educational attainment (measured in months of schooling) and citizenship acquisition. A ten percentile increase in neighborhood value-added raises educational attainment by three months and the probability of acquiring Danish citizenship by 2.5 percentage points. Results are more muted using the average neighborhood income measure, but remain significant for the educational impact.

B. *Life-Cycle Dynamics*

Figure 2 shows the life-cycle dynamics based on initial neighborhood quality at assignment, for three main outcomes: income, employment, and education. These figures reveal how gaps in outcomes form early in the life-cycle and grow gradually over time.

C. *Intermediate Outcomes*

As a first step toward understanding mechanisms, we estimate effects of assigned neighborhood quality on parental outcomes at age 30 and on neighborhood characteristics experienced between ages 0–18.

Interestingly, Table 2 shows that the impact on parental outcomes is economically meaningful and statistically significant. Using our value-added measure

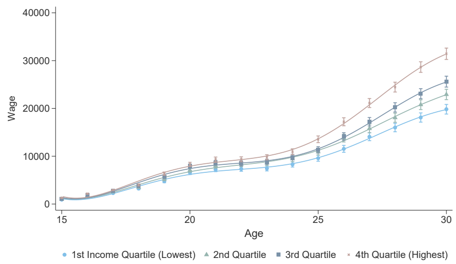
[15]Note that since β is close to 1 our observational neighborhood quality measure is closely related to the true causal effects under the hypothesis that neighborhood effects are constant across refugee children and the general population. Under this assumption, if $\beta = 1$, it would serve as an average unbiased predictor of the true causal effects. For small values of β, the observational neighborhood quality measure based on pre-tax income would exhibit little association with the true causal effect.

Table 1—Neighborhood Quality on Socio-economic Child Outcomes

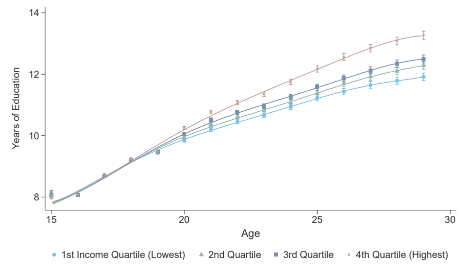
	Average Nbhd. Income		Nbhd. Value-Added	
	(1)	(2)	(3)	(4)
Panel A: Economic Integration				
Pre-Tax income	0.059	0.045	0.853	0.499
	(0.019)	(0.016)	(0.108)	(0.097)
Post-Tax Income	0.055	0.035	0.766	0.448
	(0.018)	(0.015)	(0.102)	(0.096)
Employed	0.001	0.001	0.010	0.006
	(0.000)	(0.000)	(0.002)	(0.002)
Transfers	-14.28	-14.41	-170.37	-85.08
	(5.328)	(4.741)	(34.397)	(31.578)
Panel B: Social Integration				
Months of Education	0.083	0.057	0.798	0.272
	(0.023)	(0.019)	(0.134)	(0.121)
Citizenship	0.072	0.025	0.635	0.249
	(0.031)	(0.020)	(0.138)	(0.106)
Controls	No	Yes	No	Yes

Note: Each row reports estimates from a regression of the outcome listed in the row header on assigned neighborhood quality. Columns (1)–(2) and (3)–(4) use average income and value-added measures of neighborhood quality, respectively. Odd and even columns present specifications without and with controls. Sample sizes vary slightly by outcome (see Appendix Tables S1) but are approximately 6,500 individuals. Standard errors clustered at the municipality-year level are in parentheses.

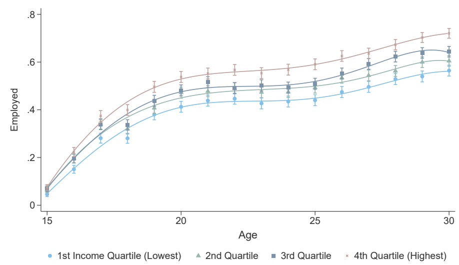
FIGURE 2. LIFE-CYCLE DYNAMICS BY INITIAL NEIGHBORHOOD QUALITY OF ASSIGNMENT



PANEL A. WAGE TRAJECTORIES



PANEL B. EDUCATION TRAJECTORIES



PANEL C. EMPLOYMENT TRAJECTORIES

Note: Panel (a) plots wage earnings by age for children of refugees assigned through the DDP. Panel (b) shows years of education, and panel (c) shows employment rates. To examine neighborhood effects over time, we divide households into quartiles based on assigned neighborhood quality. Curves are fitted using the `binsreg` command in `Stata` with a global polynomial regression. Confidence intervals are constructed using a piecewise polynomial of degree 1 with one smoothness constraint.

of neighborhood quality, we find that a one-percentile increase in neighborhood quality raises parental income rank by about .16 percentiles, but only muted effects on employment status. We also uncover significant increases in educational attainment.

Finally, we examine how assigned neighborhood quality shapes the environment children grow up in. Online Appendix Table S3 shows that assignment to higher-quality neighborhoods reduces exposure to crime and increases peer GPA, with a 1-percentile improvement in neighborhood value-added lowering crime by 0.8 percentile ranks and raising peer GPA by 0.3 percentile ranks.

Assignment to higher-quality neighborhoods substantially improves first-generation refugees' income, raising the question of whether second-generation effects operate entirely through improved parental outcomes. However, neighborhood assignment also determines children's exposure to crime, school quality (measured by peer grades), and neighborhood education levels. In Section V, we use causal mediation analysis to determine which channels—parental outcomes or neighborhood characteristics—drive the intergenerational effects.

Table 2—Neighborhood Quality on Parent Outcomes

	Average Nbhd. Income		Nbhd. Value-Added	
	(1)	(2)	(3)	(4)
Panel A: Economic Integration				
Pre-Tax income	0.025	0.021	0.414	0.158
	(0.014)	(0.010)	(0.070)	(0.059)
Post-Tax Income	0.010	0.007	0.329	0.173
	(0.013)	(0.010)	(0.064)	(0.058)
Employment Status	-0.000	0.000	0.004	0.003
	(0.000)	(0.000)	(0.002)	(0.002)
Panel B: Social Integration				
Months of Education	0.067	0.049	0.263	0.566
	(0.054)	(0.050)	(0.254)	(0.281)
Controls	No	Yes	No	Yes

Note: Each row reports estimates from a regression of the parental outcome listed in the row header on assigned neighborhood quality. Columns (1)–(2) and (3)–(4) use average income and value-added measures of neighborhood quality, respectively. Odd and even columns present specifications without and with controls. Sample sizes vary slightly by outcome (see Appendix Tables S1) but are approximately 6,500 individuals. Standard errors clustered at the municipality-year level are in parentheses.

D. Heterogeneity

We turn to evaluating heterogeneity in the effects of childhood neighborhood exposure on adult outcomes based on gender. We find that women attain greater years of education, irrespective of neighborhood quality at assignment: women obtain almost an additional year of education by age 29, relative to men. In contrast, wage trajectories and employment rates remain similar in levels across genders (see Figures S2 and S3 in Online Appendix S4.A).

Additional heterogeneity analyses appear in the Online Appendix. These analyses disaggregate outcomes based on various post-treatment variables and should be viewed as exploratory, not causal. Figure S4 shows how wages, employment, and education levels vary by the family's timing of arrival in Denmark—before birth, between ages 0 and 5, or after age 5. Outcomes decline monotonically with age at arrival, with individuals whose parents arrived before their birth achieving the best outcomes.[16]

Figures S6 and S7 further examine wage and employment dynamics by both neighborhood quality and education levels, documenting the compounding effects

[16] Timing of arrival also interacts with neighborhood quality as presented in Online Appendix Figure S5. The largest wage and education gaps appear among those born in Denmark compared to those born prior to arrival, suggesting that longer (or early) exposure to high-quality environments generates compounding benefits.

of neighborhood exposure and educational attainment on income, consistent with the findings of Nakamura, Sigurdsson and Steinsson (2022) in Iceland.

V. Testing Mechanisms: Family and Neighborhood Characteristics

In this section, we examine the channels that mediate the long-run effects of neighborhood exposure. Given the large effects of neighborhoods on parental income, we investigate the role of the family in driving the effects on children's long-term labor market income. We consider the following research question: are the effects of neighborhoods on children's long-term labor market outcomes fully explained through their effects on parental income or through neighborhood characteristics? We rely on weaker identifying assumptions, relaxing the commonly imposed conditional independence of the mediator.[17]

A. Causal Mediation Framework

We adapt Kwon and Roth's approach to studying mechanisms to our setting. For exposition purposes, suppose that we are in a perfect compliance setting and we observe $(Y, M, Z) = (Y(Z, M(Z)), M(Z), Z)$. Let $Z \in \{0, 1\}$ denote neighborhood quality at assignment, with $Z = 1$ for a high-quality neighborhood and $Z = 0$ for a low-quality neighborhood (dichotomized at the median). In this analysis, we use the average neighborhood income measure rather than the value-added measure because the latter conditions on parental income, creating potential endogeneity when testing parental income as a mediator.[18] Let $M \in \{0, 1\}$ indicate parental income measured between the ages of 0–18 of the child, with $M = 1$ for high-income parents and $M = 0$ for low-income parents (dichotomized at the median).[19] Y represents refugee children's income at ages 29–30.[20][21]

The relationship from the neighborhood quality at assignment, Z, to the outcome Y, going through the mediator, M, ($Z \to M \to Y$) reflects the sharp null of full mediation, that we investigate. A distinct effect from $Z \to Y$, if present, would violate the sharp null, indicating a direct effect of the neighborhood beyond mediation through M. This is equivalent to testing the exclusion restriction for the validity of an instrumental variable.[22] Our framework relies on the following assumptions:

[17] Alternative strategies include using instruments for M (Frölich and Huber, 2017), Difference-in-Differences (DiD) approaches (Deuchert, Huber and Schelker, 2019), and, when feasible, double randomization techniques such as parallel, paired, and cross-over designs in experimental settings (Imai, Tingley and Yamamoto, 2013).

[18] Our conclusions are robust to using the value-added measure instead.

[19] Kitagawa's (2015) test applies only to binary mediators. As a robustness check, we discretize the mediator into 10 quantile-based bins and apply the moment inequality test of Cox and Shi (2022), following Kwon and Roth (2024). Our conclusions are unchanged.

[20] The results presented in previous sections are robust to the required dichotomization of variables, as shown in Online Appendix S5.C.

[21] Following the approach in Kwon and Roth (2024), we discretize Y into 20 quantile-based bins. Our sample sizes ensures adequate cell sizes for the normal approximation underlying inference. Results are robust to using 10 bins.

[22] The proposed causal mediation relationship can be represented with a Directed Acyclic Graph (DAG)

Assumption 1 (Conditional Random Assignment). Conditional on X, neighborhood assignment Z is as good as randomly assigned: for all $z \in \{0,1\}$ and m in the support of M,

$$(4) \qquad \big(Y(z,m),\, M(z)\big) \;\perp\!\!\!\perp\; Z \,\big|\, X \quad \text{and} \quad 0 < \Pr(Z = 1 \mid X) < 1 \text{ a.s.}$$

Assumption 2 (Monotonicity). The mediator is weakly increasing in Z almost surely:

$$(5) \qquad M(1) \;\geq\; M(0) \text{ a.s.}$$

Random assignment is achieved by construction through the DDP. Monotonicity implies that being assigned to a high-income neighborhood weakly increases the likelihood of parents having a high income.

Formally, the sharp null holds if and only if $Y(z,m) = Y(m)$ almost surely for all z, m. Intuitively, if the sharp null is satisfied, then M is the sole mechanism linking Z to Y. In this case, Z serves as a valid instrument for the local average treatment effect (LATE) of M on Y, because Z affects Y only through M. By contrast, rejecting the sharp null implies that Z also has a direct effect on Y, invalidating the instrument. Using the fact that under the sharp null $Y(1,1) = Y(0,1)$, one can show that the following identifiable densities must hold:[23]

$$(6) \qquad \begin{aligned} &\Pr(y \in Y,\; M = 0 \mid Z = 0) \geq \Pr(y \in Y,\; M = 0 \mid Z = 1), \\ &\Pr(y \in Y,\; M = 1 \mid Z = 1) \geq \Pr(y \in Y,\; M = 1 \mid Z = 0). \end{aligned}$$

Accordingly, we use a variance-weighted Kolmogorov-Smirnov (KS) statistic, described in Algorithm 1 of the Online Appendix, to derive p-values.

Panel A of Table 3 reports results of the mediation analysis. We cannot reject the null of full mediation through parental income.[24] Because neighborhood assignment affects children's exposure to neighborhood characteristics (as shown in Section IV.C), we further test the null hypothesis that these characteristics fully mediate the intergenerational effects, singly and jointly. Panel B of Table 3 summarizes these neighborhood characteristics. We do reject the null for a number

in Figure S8. The path from $Z \to D$ relies on neighborhood assignment serving as a valid instrument for neighborhood of residence.

[23]The results in Kitagawa (2015) imply that these implications are sharp, but imperfect compliance in the DDP raises uncertainty about whether the property holds in practice. Kwon and Roth (2024) in our imperfect compliance setting proceeds by setting $D = Z$, under the assumption that if Z is a valid instrument for the effect of D on Y, and the sharp null hypothesis of full mediation holds, then Z affects Y only through M (see Figure S8).

[24]In Table S6, we repeat the analysis while varying the tuning parameter ξ. Specifically, we consider ξ values corresponding to $p(1-p)$ for $p = 0.005$, $p = 0.05$, and $p = 0.1$. The term $p(1-p)$ represents the variance of a Bernoulli indicator for whether an observation falls within a given interval and is used to standardize the test statistic, ensuring that differences across intervals with varying probabilities are appropriately weighted. Smaller values of ξ make the test more sensitive to small differences in the distributions, while larger values reduce sensitivity, making the test more conservative. Our results are robust to the specific choice of the tuning parameter.

of neighborhood features such as crime rate, average peer grades, and years of education and share of foreign-born. These results also alleviate concerns that Kitagawa's test has insufficient power in our data. All these results are robust to the moment inequality test of Cox and Shi (2022), which also permits conditioning on the covariates required for random assignment of refugees.[25]

Finally, we use Cox and Shi's approach to assess whether the full set of neighborhood mediators jointly accounts for the treatment effect. We reject the null of full mediation, indicating that neighborhood and school characteristics—even taken together—do not fully account for the treatment effect. We note that combining mediators mechanically reduces the power of the test Kwon and Roth (2024), which makes this rejection all the more notable. This is consistent with our main finding that parental responses are a key channel through which neighborhood quality affects child outcomes.

Table 3—Testing for Mechanisms: Parent vs. Neighborhood Mediators

Mediator	***p*-value**
Panel A. Parent Characteristics	
Parent Pre-Tax Income	.836
Panel B. Neighborhood Characteristics	
Neighborhood Average Years of Education	.001
Neighborhood Share of Foreign-Born	.000
Neighborhood Crime Rate	.000
Neighborhood Average Grades	.028

Note: Table reports *p*-values from tests of potential mediators. Panel A lists parent characteristics; Panel B lists neighborhood-level characteristics. Parental and neighborhood mediators are captured between the ages 0–18 of the child, except for neighborhood average grades which is measured between ages 15–18.

VI. Robustness to Different Neighborhood Definitions

We test the robustness of our results to different definitions of neighborhood quality, in Section S6 of the Appendix. As an alternative definition, we average across parent pre-tax, post-transfer gross income (instead of parent pre-tax, post-transfer wage income). As a second alternative, we average parent gross income pre-tax and pre-transfers. The results are reported in Table S8. We also replicate the mediation analysis using these two alternative definitions. The results are reported in Tables S9 and S10, which replicate Table 3. The mediation analysis is robust to these changes in the neighborhood definition.

[25]This approach further allows us to cluster standard errors at the municipality-year level as in our main specification.

VII. Conclusion

This paper exploits quasi-random variation in neighborhood quality among the refugee population in Denmark to study long-term neighborhood effects and underlying mechanisms. Children of refugees placed in better neighborhoods pick up about 50% of the difference in permanent residents' outcomes associated with the assigned neighborhood at around age 30. Exposure to better neighborhoods significantly improves educational attainment and reduces government transfers. Beyond economic integration, better neighborhoods foster social integration, increasing rates of naturalization by 2.5 percentage points for a 10 percentile increase in neighborhood quality.

We further introduce a novel approach in the intergenerational mobility literature for testing mechanisms and find evidence consistent with parental income playing a key role in mediating the effect of neighborhoods on children's long-run outcomes. Our findings clarify how neighborhoods can shape long-run outcomes and inform policies that foster human capital development among migrant families. Exposure to better neighborhood quality directly fosters considerable economic integration among first-generation migrants. Thus, neighborhood quality contributes to improved long-term socio-economic outcomes of refugee children by raising parental input in the human capital production function. Taken together, our findings highlight that neighborhood effects operate through fundamentally different mechanisms for parents and children.

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Online Appendix: Neighborhoods, Family and Intergenerational Mobility

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S1. Summary Statistics

Table S1—: Individual and Household Summary Statistics (in Levels)

	Mean	SD	Min	Max	N
Child Outcomes					
Wage Income	23,798	23,075	0	178,682	6589
Net Income	25,137	13,684	-18,971	135,953	6589
Government Transfers	8,082	9,487	-1,732	51,483	6589
Employment Status	0.626	0.437	0	1	6505
Citizenship	88.208	32.254	0	100	6589
Year of Birth	1,988.6	1.711	1,986	1,991	6589
College Attainment	0.333	0.471	0	1	6589
Months of Education	149.724	32.929	84	240	6038
Parent Outcomes					
Parent Wage Income	7,503	13,484	0	136,473	6456
Net Disposable Income	19,621	6,960	-447	108,646	6456

Notes: This table reports summary statistics for child and parent outcomes, in levels. Incomes are in 2015 USD.

Table S2—: Individual and Household Summary Statistics (in Ranks)

	Mean	SD	Min	Max	N
Neighborhood Attributes					
Average Wage Income	50.837	24.514	3.261	100	6583
Value-Added	42.060	3.809	25.492	63.331	6583
Average Gross Income Excl. Tr.	48.452	26.276	0.725	100	6583
Average Gross Income	47.905	26.905	0.362	100	6583
Long-Term Child Outcomes					
Wage Income	39.511	30.399	0.014	99.966	6589
Net Income	41.365	30.730	1.000	100	6589
Parent Outcomes					
Parental Wage Income	17.443	16.910	8.000	100	6456
Parental Net Income	16.600	16.424	1.000	99.000	6456
Mediators					
Parental Income Ages 0-18	8.579	10.566	0.005	95.302	6589
Parental Income Ages 0-30	9.541	13.050	0.005	98.948	6589
Share Foreign-Born	83.588	15.519	6.033	99.886	6577
Crime Rate	69.389	11.832	18.322	100	6577
Average Years of Education	79.151	16.621	1.331	100	6577
Average School Grades	53.485	21.544	1.547	100	6569

Notes: This table reports summary statistics for child and parent outcomes, in ranks. Neighborhood-level characteristics in ranks at time of refugee assignment are also reported.

S2. Data

A. Outcomes

Employment Measure: The variable *socioeconomic status* (SOCSTIL_CODE) categorizes the population's primary affiliation with the labour market according to international guidelines set by the International Labour Organisation (ILO). This is measured at the individual level, with the reference time for labor market affiliation being the end of November of the previous year. For example, the register-based labour force statistics as of January 1, 2009, are based on affiliation data from November 2008. The variable used to determine primary labour market affiliation has undergone name changes over time:

- `ARBSTIL` (1980-1993),
- `NYARB` (1994-1995),
- `SOCSTIL` (1996-2008).

These changes reflect shifts in the level of detail regarding the population's employment status. In 1994, `NYARB` introduced a more detailed classification for persons outside the labour force, and in 1996, `SOCSTIL` replaced `NYARB` to refine the employee category.

The labour force is divided into three primary groups, based on the variable `NOVPRIO=1` (November priority):

1) **Employed**: ARBSTIL/NYARB = 11-37, SOCSTIL = 115-135
2) **Unemployed**: ARBSTIL/NYARB = 40, SOCSTIL = 200-201
3) **Persons outside the labour force**: ARBSTIL = 50-90, NYARB/SOCSTIL = 310-400

The method for calculating these groups remained consistent across three periods: 1980-2001, 2002-2007, and 2008 onwards.[1]

B. Neighborhood Characteristics Measures.

School peer quality: Average student GPA at the end of middle school (around age 15 in Denmark) in the neighborhood. Values are standardized to have mean 0 and standard deviation 1 across municipalities within each year.

Crime rate: Length of prison sentence in days averaged at the neighborhood level[2].

[1] **Source:** Statistics Denmark, "Socioeconomic Status (SOCSTIL Code)"https://www.dst.dk/da/TilSalg/data-til-forskning/generelt-om-data/dokumentation-af-data/hoejkvalitetsvariable/befolkningens-tilknytning-til-arbejdsmarkedet--ras-/socstil-kode

[2] **Source:** Statistics Denmark, "AFG_UBSTRFLG"https://www.dst.dk/da/Statistik/dokumentation/Times/kriminalstatistik/afg-ubstrflg

Average years of education: Mean years of education at household-level, averaged at the neighborhood, calculated using `HFPRIA` (standard time for highest completed education). This captures the overall educational attainment of the neighborhood population.

Share of migrants: Proportion of neighborhood residents who are migrants. This measure is calculated using country-of-origin information from population registers (`IELAND`).

C. Timing of Variable Measurement

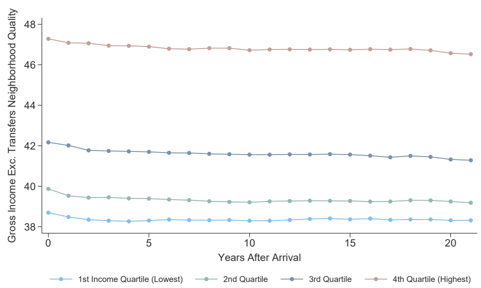
Child Outcomes: Children's income, employment, government transfers and citizenship are observed at ages 29 and 30. Individual education is measured at age 29. Parental education is measured when the child is age 29.

Parent Outcomes: Parental income is averaged over the period in which the child is ages 24–30, when it acts as an outcome. Parental employment is measured when the child is ages 29 and 30.

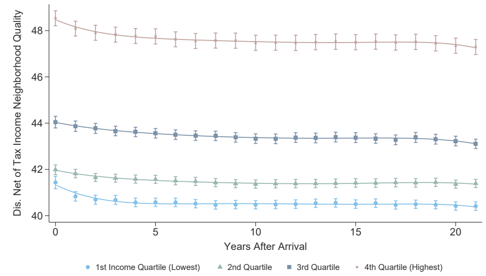
Mediators: Neighborhood quality measures are captured when the child is ages 0–18, except for school grades which can only be observed from 2001 onward. Thus, average grades at neighborhood quality is captured between ages 15-18. When parental income is considered as a mediator, it is captured as the average income between ages 0–18 and 0–30 to closely approximate permanent income.

S3. Sorting Patterns

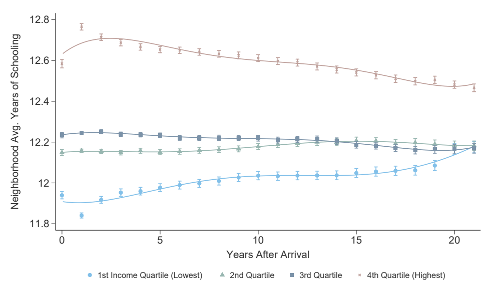
Figure S1. : Sorting of Individuals by Neighborhood Quality of Assignment



(a) Neighborhood Quality: Gross Income (Excluding Tax and Transfers)



(b) Neighborhood Quality: Net Income



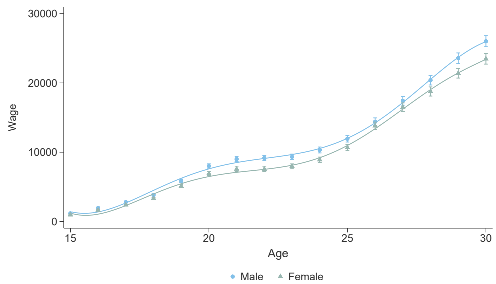
(c) Neighborhood Avg. Years of Education

Note: Sorting patterns of individuals by quartiles of initial quality of assignment. Panel (a) uses our value-added neighborhood quality measure based on gross income, as defined in Equation (2), while panel (b) is defined similarly based on net income. Panels (c) refers to average education at the neighborhood level. Using the `binsreg` command in `Stata`, a global polynomial regression model was fitted for plotting. Confidence intervals were constructed using a piecewise polynomial of degree 1 with one smoothness constraint.

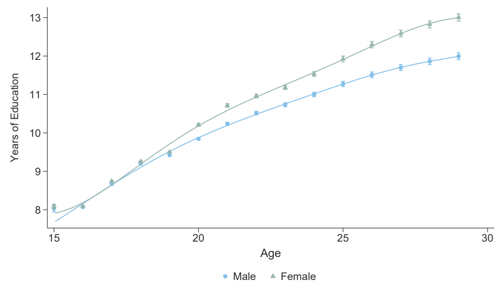
S4. Life-Cycle Dynamics

A. Gender Dynamics

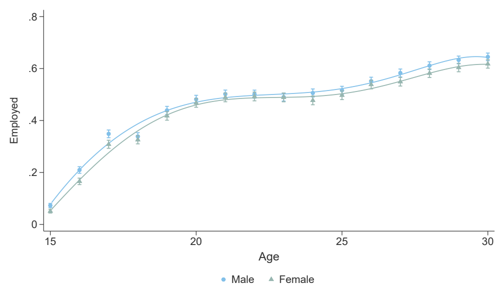
Figure S2. : Life-Cycle Dynamics by Gender



(a) Wage Dynamics by Gender



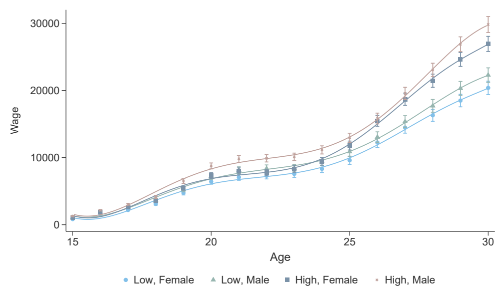
(b) Education Dynamics by Gender



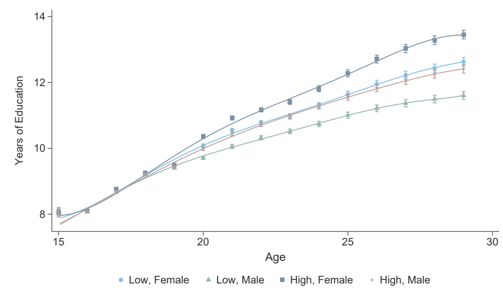
(c) Employment Dynamics by Gender

Note: Panel (a) presents the wages of the children of refugees treated during the tenure of the DPP by gender. Panel (b) shows years of education, while Panel (c) displays employment. Using the `binsreg` command in `Stata`, a global polynomial regression model was fitted for plotting purposes. Confidence intervals were constructed using a piecewise polynomial of degree 1 with one smoothness constraint.

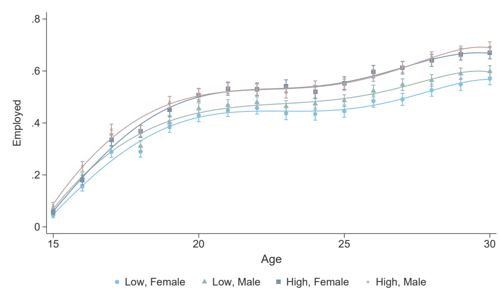
Figure S3. : Life-Cycle Dynamics by Gender and Neighborhood Quality at Assignment



(a) Wage Trajectories by Gender and Neighborhood Quality



(b) Educational Trajectories by Gender and Neighborhood Quality

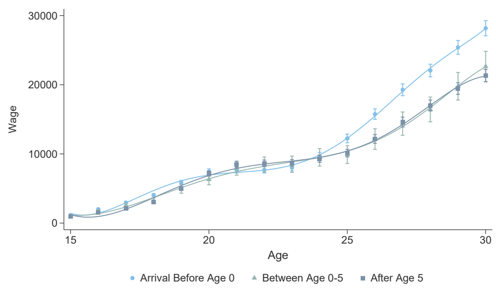


(c) Employment Trajectories by Gender and Neighborhood Quality

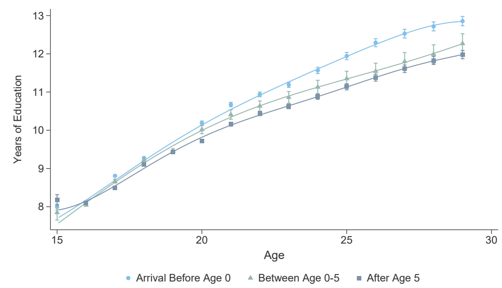
Note: Panel (a) presents the wages of the children of refugees treated during the tenure of the DPP, categorized by gender and neighborhood quality at arrival (low vs. high, based on a median split of pre-tax income). Panel (b) shows years of education, while Panel (c) displays employment status. Using the `binsreg` command in `Stata`, a global polynomial regression model was fitted for plotting purposes. Confidence intervals were constructed using a piecewise polynomial of degree 1 with one smoothness constraint.

B. *Additional Heterogeneity Analyses across Life-Cycle*

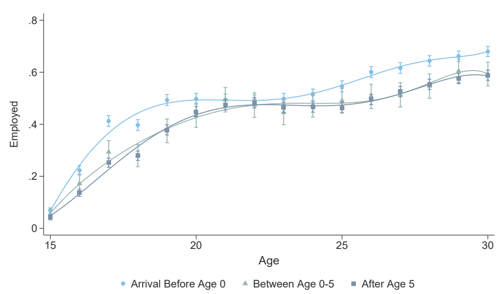
Figure S4. : Life-Cycle Dynamics by Timing of Arrival



(a) Wage Dynamics by Timing of Arrival



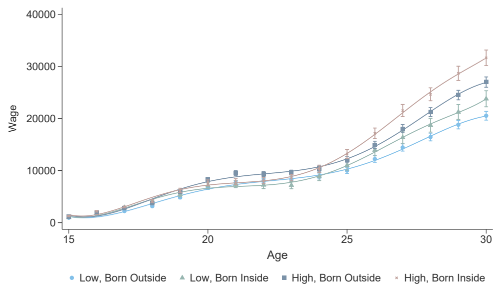
(b) Education Dynamics by Timing of Arrival



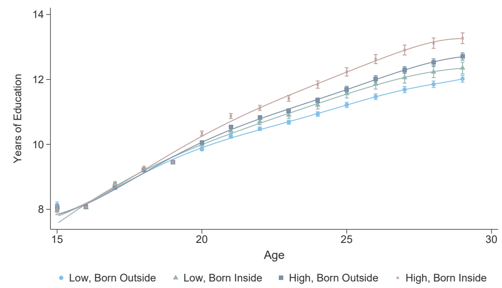
(c) Employment Dynamics by Timing of Arrival

Note: Panel (a) presents the wages of the children of refugees treated during the tenure of the DPP, categorized by timing of arrival in Denmark—those born in Denmark (arrival before age 0) versus those who arrived between ages 1 and 5 or later. Panel (b) shows years of education, while Panel (c) displays employment. Using the `binsreg` command in `Stata`, a global polynomial regression model was fitted for plotting purposes. Confidence intervals were constructed using a piecewise polynomial of degree 1 with one smoothness constraint.

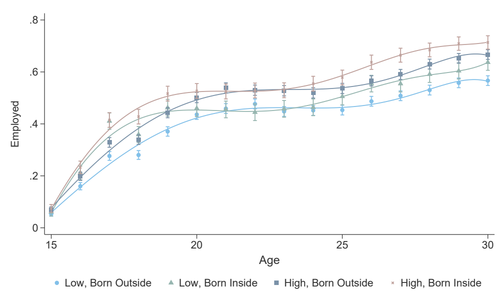
Figure S5. : Life-Cycle Dynamics by Timing of Arrival and Neighborhood Quality at Assignment



(a) Wage Trajectories by Timing of Arrival and Neighborhood Quality



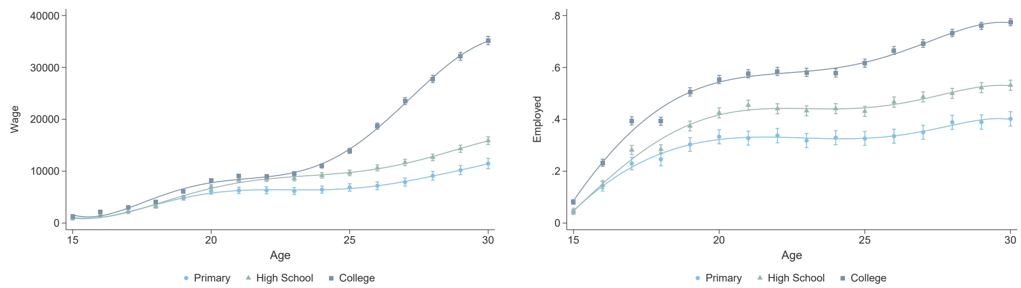
(b) Educational Attainment by Timing of Arrival and Neighborhood Quality



(c) Employment Trajectories by Timing of Arrival and Neighborhood Quality

Note: Panel (a) shows average wages of children of refugees treated under the DPP, disaggregated by timing of arrival—those born in Denmark (arrival before age 1) versus those who arrived after birth—and by neighborhood quality at arrival (low vs. high, based on a median split of pre-tax income). Panel (b) reports years of education, and Panel (c) shows employment status. Using the `binsreg` command in `Stata`, a global polynomial regression model was fitted for plotting purposes. Confidence intervals were constructed using a piecewise polynomial of degree 1 with one smoothness constraint.

Figure S6. : Life-Cycle Dynamics by Different Levels of Schooling

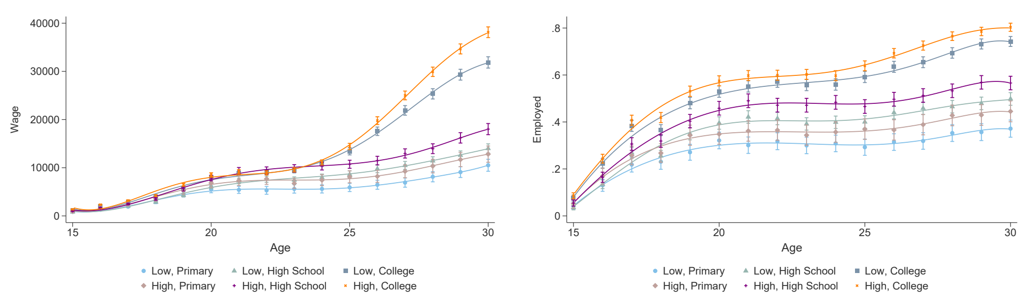


(a) Wage Dynamics by Schooling Level

(b) Employment Dynamics by Schooling Level

Note: Panel (a) presents the wages of the children of refugees treated during the tenure of the DPP, categorized by different levels of schooling. Panel (b) shows years in employment. Using the `binsreg` command in `Stata`, a global polynomial regression model was fitted for plotting purposes. Confidence intervals were constructed using a piecewise polynomial of degree 1 with one smoothness constraint.

Figure S7. : Life-Cycle Dynamics by Education Level and Neighborhood Quality at Assignment



(a) Wage Trajectories by Education Level and Initial Neighborhood Quality

(b) Employment Trajectories by Education Level and Initial Neighborhood Quality

Note: Panel (a) presents average wages and Panel (b) reports employment status for children of refugees treated under the DDP. Results are disaggregated by the child's education level—Primary, High School, and College—and neighborhood quality at assignment, where neighborhood quality is classified as low or high based on a median split of pre-tax income. Using the `binsreg` command in `Stata`, a global polynomial regression model was fitted for plotting purposes. Confidence intervals were constructed using a piecewise polynomial of degree 1 with one smoothness constraint.

S5. Testing Mechanisms

A. Directed Acyclic Graph

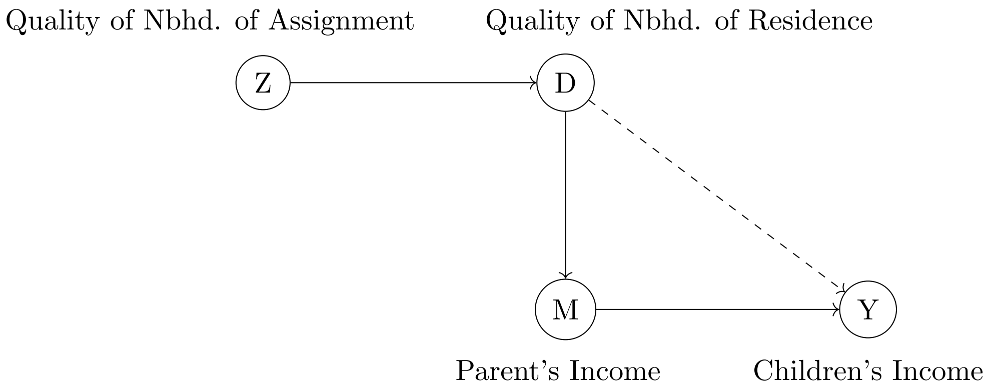


Figure S8. : Directed Acyclic Graph illustrating the proposed structural model

Note: Directed Acyclic Graph illustrating the proposed structural model, assuming the validity of instrument Z, neighborhood of assignment, and the null hypothesis of sharp mediation.

B. Estimates of Neighborhood Effects on Mediators

Table S3—: Neighborhood ITT Estimates: Impact of Assigned Neighborhood on Neighborhood Attributes

	Average Nbhd. Income (1)	Nbhd. Value-Added (2)
Neighborhood Share of Migrants	-0.020	-0.944
	(0.024)	(0.102)
Average Neighborhood Crime	-0.094	-0.818
	(0.017)	(0.071)
Average Neighborhood Education	0.061	-0.691
	(0.025)	(0.112)
Average Neighborhood GPA	0.056	0.307
	(0.028)	(0.132)
Controls	Yes	Yes

Notes: This table reports the effects of assigned neighborhood quality on neighborhood characteristics experienced during childhood (ages 0–18). Column headers indicate the neighborhood quality measure used: average income (columns 1) or value-added (columns 2). Sample sizes vary slightly by outcome (Appendix Table S1) but are approximately 6,500 individuals. Standard errors are clustered at the municipality-year level.

C. *Estimates of Neighborhood Effects on Children's Long-term Outcomes Based on Dichotomized Variables*

Table S4—: Estimates: Neighborhood Effects on Children's Long-term Outcomes Using Treatment Dummy

	Average Nbhd. Income		Nbhd. Value-Added	
	(1)	(2)	(3)	(4)
Panel A: Economic Integration				
Pre-Tax income	0.300	0.221	0.588	0.346
	(0.093)	(0.074)	(0.080)	(0.073)
Post-Tax Income	0.219	0.119	0.473	0.253
	(0.083)	(0.071)	(0.075)	(0.073)
Employed	0.038	0.028	0.078	0.046
	(0.013)	(0.011)	(0.011)	(0.011)
Transfers	-846.295	-763.162	-1.6e+03	-1.0e+03
	(311.287)	(258.063)	(260.855)	(241.837)
Panel B: Social Integration				
Months of Education	4.397	2.734	6.371	2.594
	(1.188)	(0.929)	(0.975)	(0.906)
Citizenship	3.240	0.489	5.015	1.852
	(1.519)	(0.918)	(1.246)	(0.867)
Controls	No	Yes	No	Yes

Notes: This table shows results for the ITT specification, focusing on child outcomes. Treatments and outcomes are dichotomized. Each row reports estimates from a regression of the outcome listed in the row header on assigned neighborhood quality. Columns (1)–(2) and (3)–(4) use average income and value-added measures of neighborhood quality, respectively. Odd and even columns present specifications without and with controls. Sample sizes vary slightly by outcome (see Appendix Tables S1) but are approximately 6,500 individuals. Standard errors clustered at the municipality-year level are in parentheses.

Table S5—: Estimates: Neighborhood Effects on Parents Using Binary Treatment Dummy

	Average Nbhd. Income		Nbhd. Value-Added	
	(1)	(2)	(3)	(4)
Panel A: Economic Integration				
Pre-Tax income	1.434	0.613	3.675	1.630
	(0.677)	(0.533)	(0.518)	(0.431)
Post-Tax Income	0.925	0.265	2.516	1.152
	(0.601)	(0.513)	(0.456)	(0.418)
Employed	-0.003	0.001	0.034	0.024
	(0.015)	(0.013)	(0.013)	(0.011)
Panel B: Social Integration				
Months of Education	1.587	1.213	0.290	4.482
	(2.614)	(2.413)	(2.129)	(2.224)
Controls	No	Yes	No	Yes

Notes: This table shows results for the ITT specification, focusing on parental outcomes. Treatments and outcomes are dichotomized. Each row reports estimates from a regression of the outcome listed in the row header on assigned neighborhood quality. Columns (1)–(2) and (3)–(4) use average income and value-added measures of neighborhood quality, respectively. Odd and even columns present specifications without and with controls. Sample sizes vary slightly by outcome (see Appendix Tables S1) but are approximately 6,500 individuals. Standard errors clustered at the municipality-year level are in parentheses.

D. Algorithm for Kitagawa's test

Algorithm 1 Kitagawa (2015) (Algorithm 2.1)

Let P and Q be the conditional probability distributions of (Y, M) given $Z = 1$ and $Z = 0$, respectively. Let ξ be a tuning parameter. In our results, we do the bootstrap 500 times. The different steps for the test, suggested by Kwon and Roth (2024), are summarized below.

1: **Step 1:** Sample (Y_i^*, M_i^*), $i = 1, \ldots, k$, randomly with replacement from $H_N = \hat{\lambda} P_k + (1 - \hat{\lambda}) Q_n$ and construct the empirical distribution P_k^*.

2: Similarly, sample (Y_j^*, M_j^*), $j = 1, \ldots, n$, randomly with replacement from H_N and construct the empirical distribution Q_n^*.

3: **Step 2:** Calculate a bootstrap realization of the test statistic:

$$T_N^* = \left(\frac{kn}{N}\right)^{1/2} \max \left\{ \begin{array}{c} \sup\limits_{-\infty \le y \le y' \le \infty} \left\{ \dfrac{|Q_n^*([y, y'], 1) - P_k^*([y, y'], 1)|}{\xi \vee \sigma_{P_m^*, Q_n^*}([y, y'], 1)} \right\}, \\ \sup\limits_{-\infty \le y \le y' \le \infty} \left\{ \dfrac{|P_k^*([y, y'], 0) - Q_n^*([y, y'], 0)|}{\xi \vee \sigma_{P_k^*, Q_n^*}([y, y'], 0)} \right\} \end{array} \right\}$$

where

$$\sigma^2_{P_k^*, Q_n^*}([y, y'], z) = (1 - \hat{\lambda}) P_k^*([y, y'], z)(1 - P_k^*([y, y'], z)) + \hat{\lambda} Q_n^*([y, y'], z)(1 - Q_n^*([y, y'], z)).$$

4: **Step 3:** Repeat Steps 1 and 2 many times to obtain the empirical distribution of T_N^*. For a nominal level $\alpha \in (0, 1/2)$, compute the bootstrap critical value $c_{N,1-\alpha}$ as the $(1 - \alpha)$ quantile of the empirical distribution.

5: **Step 4:** Reject the null hypothesis (1.1) if $T_N > c_{N,1-\alpha}$. The bootstrap p-value is the proportion of repetitions such that $T_N^* > T_N$.

E. Sensitivity Analyses of Kitagawa's Tests

Table S6—: P-values from the Kitagawa test for different values of the tuning parameter

	$\sqrt{0.005(1-0.005)}$	$\sqrt{0.05(1-0.05)}$	$\sqrt{0.1(1-0.1)}$	1
Panel A. Parent Characteristics				
Pre-Tax income	0.824	0.886	0.834	0.924
Panel B. Neighborhood Characteristics				
Average Years of Education	0.004	0.004	0.000	0.000
Share Foreign-Born	0.000	0.000	0.000	0.000
Crime Rate	0.000	0.000	0.000	0.000
Average Grades	0.026	0.006	0.000	0.012

Notes: The table reports p-values from the Kitagawa test under different choices of the tuning parameter for parental income and neighborhood characteristics as the mediators, defined as $\sqrt{p(1-p)}$ with $p \in \{0.005, 0.05, 0.1, 1\}$.

S6. Robustness to Different Definitions

Table S7—: Child ITT Estimates: Alternative Neighborhood Quality Definitions

	Nbhd. Pre-Tr. Income		Nbhd. Post-Tr. Income	
	(1)	(2)	(3)	(4)
Panel A: Economic Integr.				
Pre-Tax income	0.063	0.046	0.063	0.044
	(0.017)	(0.015)	(0.017)	(0.014)
Post-Tax Income	0.054	0.034	0.055	0.031
	(0.017)	(0.014)	(0.016)	(0.014)
Employed	0.001	0.001	0.001	0.001
	(0.000)	(0.000)	(0.000)	(0.000)
Transfers	-15.313	-14.154	-15.533	-14.136
	(4.958)	(4.467)	(4.790)	(4.383)
Panel B: Social Integr.				
Months of Education	0.080	0.052	0.083	0.051
	(0.022)	(0.018)	(0.021)	(0.017)
Citizenship	0.067	0.024	0.077	0.027
	(0.030)	(0.019)	(0.030)	(0.019)
Controls	No	Yes	No	Yes

Notes: This table shows results for the ITT specification, focusing on child outcomes. In columns (1)-(2), neighborhood quality is defined as the average gross income pre-tax and pre-transfers at the neighborhood-level at assignment. In columns (3)-(4), neighborhood quality is defined as the average gross income pre-tax and post-transfers at the neighborhood-level at assignment. Each row reports estimates from a regression of the outcome listed in the row header on assigned neighborhood quality. Odd and even columns present specifications without and with controls. Sample sizes vary slightly by outcome (see Appendix Tables S1) but are approximately 6,500 individuals. Standard errors clustered at the municipality-year level are in parentheses.

Table S8—: Parent ITT Estimates: Alternative Neighborhood Quality Definitions

	Nbhd. Pre-Tr. Income		Nbhd. Post-Tr. Income	
	(1)	(2)	(3)	(4)
Panel A: Economic Integr.				
Pre-Tax income	0.021	0.017	0.025	0.019
	(0.013)	(0.009)	(0.013)	(0.009)
Post-Tax Income	0.006	0.004	0.008	0.005
	(0.013)	(0.009)	(0.012)	(0.009)
Employed	-0.000	0.000	-0.000	0.000
	(0.000)	(0.000)	(0.000)	(0.000)
Panel B: Social Integr.				
Months of Education	0.058	0.060	0.052	0.058
	(0.052)	(0.046)	(0.052)	(0.046)
Controls	No	Yes	No	Yes

Notes: This table shows results for the ITT specification, focusing on parent outcomes. In columns (1)-(2), neighborhood quality is defined as the average gross income pre-tax and pre-transfers at the neighborhood-level at assignment. In columns (3)-(4), neighborhood quality is defined as the average gross income pre-tax and post-transfers at the neighborhood-level at assignment. Each row reports estimates from a regression of the outcome listed in the row header on assigned neighborhood quality. Odd and even columns present specifications without and with controls. Sample sizes vary slightly by outcome (see Appendix Tables S1) but are approximately 6,500 individuals. Standard errors clustered at the municipality-year level are in parentheses.

Table S9—: Testing for Mechanisms with Alternative Neighborhood Quality Definition: Parent vs. Neighborhood Mediators

Mediator	p**-value**
Panel A. Parent Characteristics	
Parent Pre-Tax Income	.98
Panel B. Neighborhood Characteristics	
Neighborhood Average Years of Education	.000
Neighborhood Share of Foreign-Born	.000
Neighborhood Crime Rate	.000
Neighborhood Average Grades	.000

Table reports p-values from tests of potential mediators using an alternative measure for assigned neighborhood quality. Neighborhood quality is defined as the average gross income pre-tax and pre-transfers at the neighborhood-level at assignment. Parent income measures are captured between ages 0–18 of the child. Panel A lists parent characteristics; Panel B lists neighborhood-level characteristics.

Table S10—: Testing for Mechanisms with Alternative Neighborhood Quality Definition: Parent vs. Neighborhood Mediators

Mediator	p-value
Panel A. Parent Characteristics	
Parent Pre-Tax Income	.92
Panel B. Neighborhood Characteristics	
Neighborhood Average Years of Education	.000
Neighborhood Share of Foreign-Born	.000
Neighborhood Crime Rate	.000
Neighborhood Average Grades	.000

Table reports p-values from tests of potential mediators using an alternative measure for assigned neighborhood quality. Neighborhood quality is defined as the average gross income pre-tax and post-transfers at the neighborhood-level at assignment. Parent income measures are captured between ages 0–18 of the child. Panel A lists parent characteristics; Panel B lists neighborhood-level characteristics.

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Kwon, Soonwoo and Jonathan Roth, "Testing Mechanisms," *arXiv preprint arXiv:2404.11739*, 2024.

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